

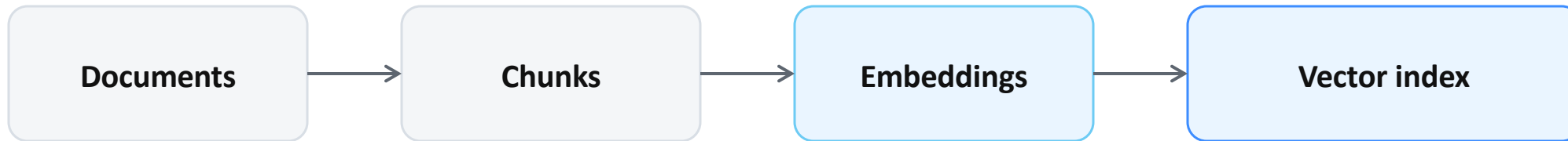
When RAG Finds the Right Document - but Not the Answer



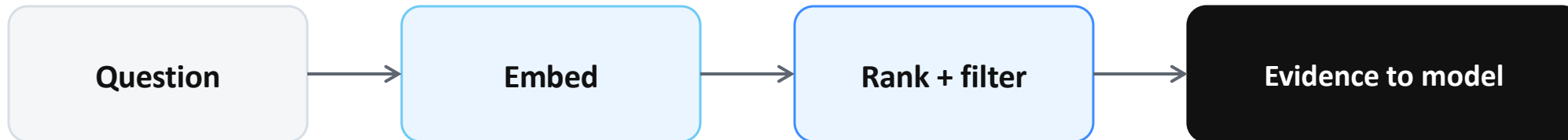
A retrieval failure, an agentic fix, and the lesson behind both.

RAG has two paths - prepare knowledge once, retrieve it many times.

INGESTION



QUERY



A useful lab begins with a prediction, then reveals the evidence.

QUESTION

What is the maximum daily meal allowance during business travel?

0.49

top cosine score

>

0.45

frozen threshold

-

>

travel_policy.txt

retrieved - but amount unspecified

The retriever answered 'where should I look?' - not 'is the requested fact present?'

HELD-OUT CONFUSION COUNTS

3

true positive

1

false positive

2

true negative

0

false negative

75%

precision

100%

recall

Raising 0.45 above 0.49 would fix this row - and leak the answer into the model choice.

TEMPTATION

**Raise the threshold
above 0.49**

One failure disappears.
Confidence looks better.

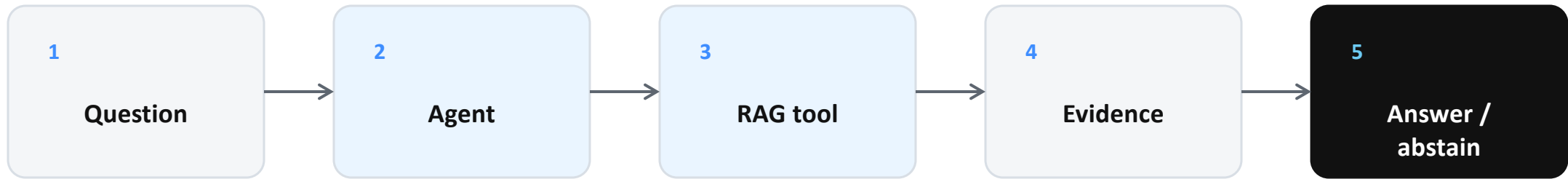
DISCIPLINE

Freeze -> test -> diagnose

Expand the labeled set. Add sufficiency checks.
Recalibrate only on validation data.

This is the RAG equivalent of backtesting without peeking.

The tool finds evidence; the agent decides whether that evidence can support an answer.



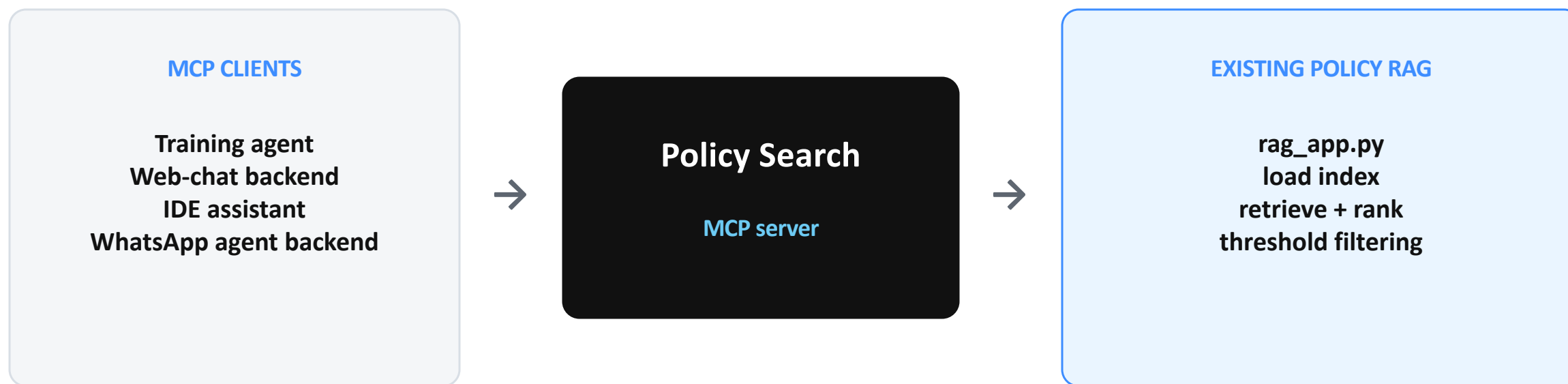
“The travel policy covers reimbursable meal expenses, but it does not specify a maximum daily amount.”

Each step gives learners a reason to care about the next.



The method works for RAG, agents, MCP, APIs, and live troubleshooting.

The clients change; the existing RAG logic stays in one place.



MCP standardizes discovery, schemas, and tool calls - it does not replace the RAG.